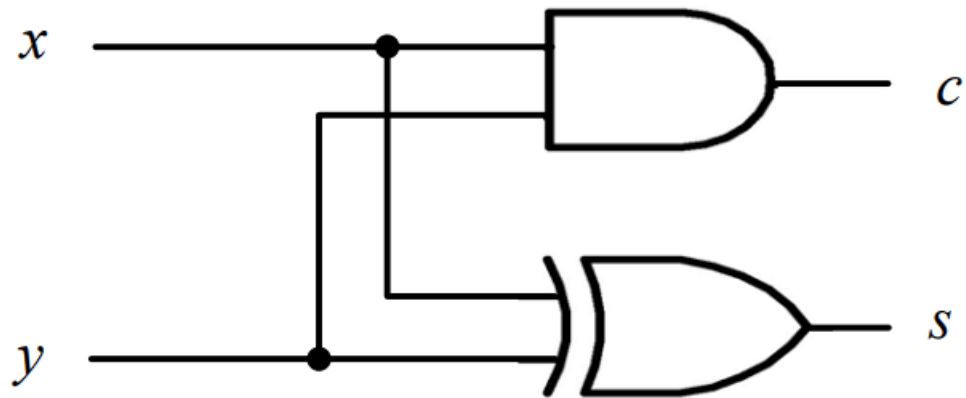


Digital Logic Design – Lab 1

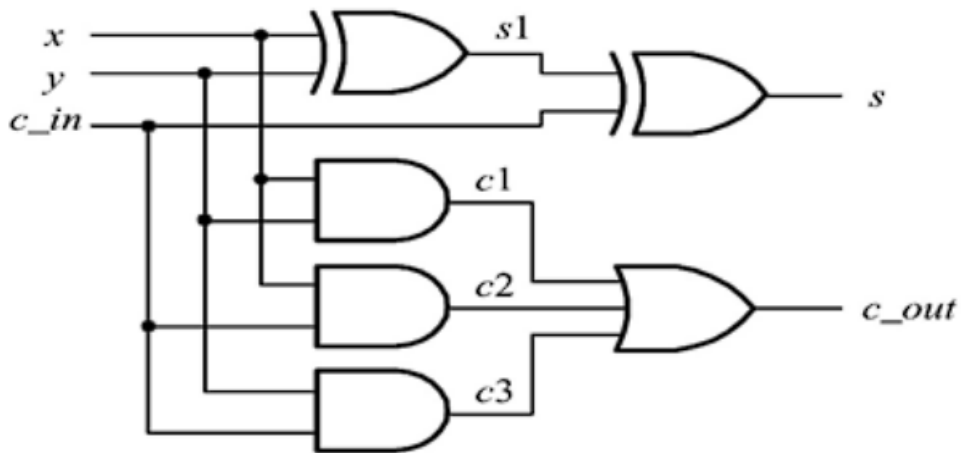
1. **Topic:** Half Adder, and Full Adder

2. **Design structure:**

- Half Adder



- Full Adder



3. Code and Test bench:

- Half Adder

```
// Half Adder circuit using Gate Level modeling
module half_adder(x, y, c, s);
    // input ports
    input x, y;

    // output ports
    output c, s;

    // Gate level implementation of Half Adder
    and AND(c, x, y);
    xor XOR(s, x, y);
endmodule

module half_adder_sim();
    // inputs
    reg x, y;

    // outputs
    wire c, s;

    // Instantiate the Unit Under Test
    half_adder uut(.x(x), .y(y), .c(c), .s(s));

    initial begin
        // Initialize inputs
        x = 0; y = 0;
        #100; // wait 100ns

        x = 0; y = 1;
        #100;

        x = 1; y = 0;
        #100;

        x = 1; y = 1;
        #100;
    end
endmodule
```

- Full Adder

```
// Full Adder circuit using Gate Level modeling
module full_adder(x, y, c_in, s, c_out);
    // input ports
    input x, y, c_in;

    // output ports
    output s, c_out;

    // wires for intermediate connections
    wire s1, c1, c2, c3;

    // Gate Level implementation of Full Adder
    xor XOR_1 (s1, x, y);
    xor XOR_FINAL (s, s1, c_in);

    and AND_1 (c1, x, y);
    and AND_2 (c2, x, c_in);
    and AND_3 (c3, y, c_in);
    or OR_FINAL (c_out, c1, c2, c3);
endmodule
```

```

module full_adder_sim();
    // inputs
    reg x, y, c_in;

    // outputs
    wire s, c_out;

    // Instantiate the Unit Under Test
    full_adder uut(.x(x), .y(y), .c_in(c_in), .s(s), .c_out(c_out));

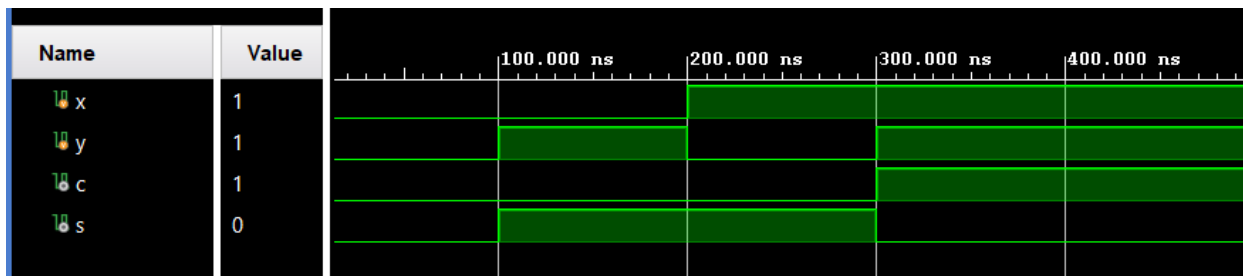
    initial begin
        // Initialize inputs
        x = 0; y = 0; c_in = 0;
        #100; // wait 100ns
        x = 0; y = 0; c_in = 1;
        #100;
        x = 0; y = 1; c_in = 0;
        #100;
        x = 0; y = 1; c_in = 1;
        #100;

        x = 1; y = 0; c_in = 0;
        #100;
        x = 1; y = 0; c_in = 1;
        #100;
        x = 1; y = 1; c_in = 0;
        #100;
        x = 1; y = 1; c_in = 1;
        #100;
    end
endmodule

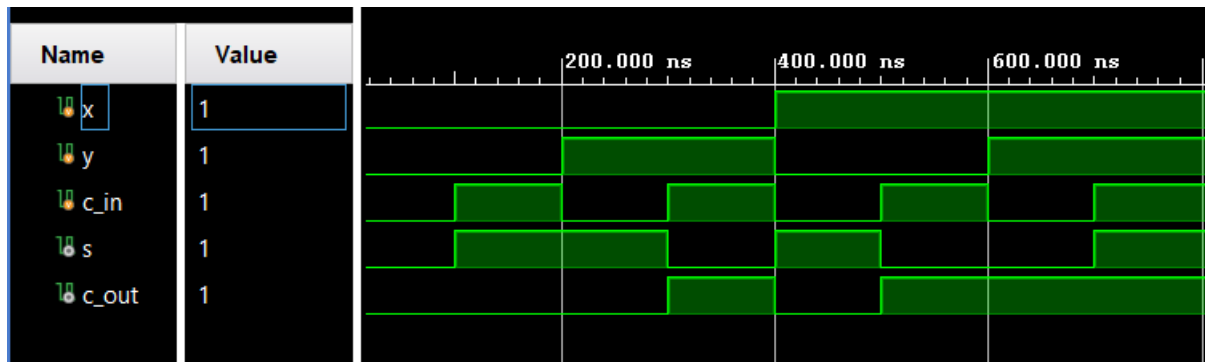
```

4. Result (Simulation waves):

- Half Adder



- Full Adder



5. Reflection

In this lab, we designed a basic circuit for a half adder and full adder, by programming it in the simplest way, gate level modeling. It was a good first experience implementing such circuits and visualizing their results on the simulation of waves